

COMPOSITE MATERIALS LAB

IV B. Tech. I Semester: MECHANICAL ENGINEERING

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6ME42	PCC	0	0	2	1	40	60	100
COURSE OVERVIEW: Composite Material Lab is intended to teach the basics of composites manufacturing including polymer matrix composites and metal matrix composites. Experiments are framed to guide the learning process and impart technical knowledge in selecting the matrix and reinforcement material for fabricating the composites. COURSE OUTCOMES: At the end of the course, the student should be able to: 1. Fabricate the metal and polymer matrix composites. 2. Determine the mechanical properties of laminated composites. 3. Determine the voids and flaws that may present in the composite materials. 4. Prepare the metal matrix composites using stir casting machine. 5. Find the mechanical properties of MMCs.								
LIST OF EXPERIMENTS: 1. Fabrication of a laminated composite using hand lay-up process. 2. Study on the flexural properties of laminated composites using three point bend fixture. 3. Determining the Barcol Hardness of Laminated Composites. 4. Detecting the Voids and Flaws in Laminated Composites using Ultrasonic Flaw Detector. 5. Preparation of Aluminium Metal Matrix composites using Stir Casting Machine. 6. Determining the tensile properties of MMCs using UTM. 7. Determining the hardness of MMCs using Rockwell Hardness Tester. 8. Study on the flexural properties of MMCs using three point bend fixture. 9. Study the Impact properties of MMCs using impact test machine. 10. Detecting the Voids and Flaws in MMCs using Ultrasonic Flaw Detector								